

ActInf GuestStream #034.1 ~ Avel Guénin-Carlut ~ "Physics of Creation"

GuestStream | Avel Guénin-Carlut ~ | 2:01:20

Good luck.

Hello, I'm ready.

I'm ready.

Yes, we are in.

It is January 23rd, 2023.

We're here in Active Inference Guest Stream number 34.1
with Avel, Gwinn, Carlou.

We're going to have a one hour presentation,
Physics of Creation,

Can the Free Energy Principle

Ground the Construction of Physical State Spaces,
followed by about an hour discussion.

So if you're watching live,
please feel free to add questions in the live chat
and we'll also have some guests.

So thank you, Avel, for joining
and off to you for the presentation.

Hello.

So this is a complete presentation.

I will first introduce myself and the work.

I am Avel Gwinn, Carlou.

I have a problem in Physics.

I'm working in cognitive science under
version of Andy Clark.

This work is, essentially speaking,
the product of Kairos Research,
which is a laboratory which myself,
Alexei Rosansky, who is present here today,
and other people are funded to
investigate issues of events to political
through political evolution and cognitive science.

I will be on the physics of creation.

So creation is an atypical term, let us say.

It will be contextualized in presentation,
but let us first define that it corresponds to the construction of physical possibilities,

physical spaces to talk in a more technical language.

And the original question that motivated this research is whether or not the Frederick Principle, a piece of math that emerges in the cognitive science, which we will get to very soon, is capable of representing this process of creation.

So first, I have tried a first run of this without getting into the technicals.

It did not work.

So I will address the technicals first.

We will see some concepts in physics, what they do, how they work, how they relate to what people typically do in physics.

And first, I'd like to take a look at how physics explain.

So physics explain with laws.

They think that in nature, you have laws.

And those laws are basically formal statements that constrain the possible states of a system.

And typically, those are entailing laws from which you can derive logically what the system will do.

An example of natural entailing law is Maxwell's equation of Electromagnetics.

It is a different Maxwell than the one that is here.

So what you see is four bits of vectorial calculus that will tell you how two fields are collectively the electromagnetic fields evolve with regard to another.

So this is a formal constraint on what E and B can do.

So the equality express a constraint by definition.

And those can be used to derive basically fully E and B given boundary conditions, which is how you do things in electromagnetism.

So what is of relevance here is that any of those laws presupposes basically that I can define E as a vectorial field.

So it presupposes what is called a state space, which is mathematical space that represents the set of possible states.

Before I state ΔE and ΔB is equal to this and this,

I have to state that E exists and E can take such and such states.

In that case, it is a vectorial space in three dimensions.

And this set space basically does by construction to capture any difference that make a difference about the system that is targeted.

For any change in E to be basically computable by math theory, it has to be represented in the set space.

So by construction, I want the set space to categorify every variable of interest in my system, given the scope and measurement capabilities of my theory.

You have underlying notion that is basically implicitly underlying the notion of flows and set space, that is the notion of symmetry.

I cannot get into the gory detail here, but I will try to be as complete as possible.

We specifically understand physical theories outside, specifically relativity theory and some weird quantum

stuff,

to derive from symmetries.

Symmetries are, I will say it out loud,

algebraic group of transformation of other set space that do not affect a system's dynamics as presented as Laplacian.

You do not have to understand what I just said. What you have to understand is that there are ways in which I can transform the system,

which does not change what it is.

An example in the symmetries, the ceiling of the Lotfella Mosque in Intfau,

you see that you have basically the same pattern that is repeated again and again and again and again radially.

So if I make a small rotation in either direction, it will not change a thing.

And if I repeat it and repeat it and repeat it and repeat it and repeat it and repeat it,

I will still change my system. Formally I will apply transformation, but it does not change

basically what observable properties the system have.

And this is what a symmetry is. It is a set of transformation that is generated by a subset of smaller transformation and that does not change what the system observably is.

And because of this, symmetries of a system, they basically constrain what difference can make a difference about the system. So implicitly when I formulate a law or set space, I make a statement about the underlying symmetries. I make the statement that basically everything that is not in the system, that is not represented as part of the system, is something that is symmetric, that the system is symmetric with regard to that on which any transformation will not affect the system trajectory, at least not in a way that is relevant to my theory.

It works for physics because it looks like in physics we have a few symmetries that are everywhere verified, that are fundamental in some sense. It doesn't work so well in life and mind because we don't have that.

To be specific, living systems have a structure, I do not think it should be controversial, and they have constitutive symmetries. They have changes that will not affect them. If I switch all of the carbon atoms in my body,

I scramble them between each other while keeping the relation to the rest, it did not change a thing.

So it is very trivial in the sense of symmetry, but basically every structure that is me entails symmetries, entail some transformations that are possible that do not affect the structure and other that will affect it.

And the structure it is not a given, like symmetries are considered to be in physics.

It is actively reconstructed by the activity of me. If I cut myself, if I fall, if my bones break,

they will regrow basically. You will have circular tissue that will occur and that will make it okay again.

And it will change what structure is my body. Most importantly, I was not born that way, like big and with a bird and with 80 kilos of meat.

I grew. And I grew from one single cell. And this single cell just created an anatomical structure,

apparently ex nihilo, from basically a genetic code in the context. And not only that, but life in itself evolves in

a way that builds structures,

that builds symmetries. And all of this means that you can't set a state space that is the space of life in general,

of any living system. The symmetries that are in the life system, they change through time,

and they change by the own activity of this system. So any living system constructs set spaces and to

account for it, we need a theory of space construction. This is the motivation of the physics of creation

line of work. And this is basically the, I did not frame it as a question, it is the question that we

will actually need to keep in mind in this discussion. So Daniel, do we have any question yet?

Anyone? Okay, no technical issues that I should know about.

Okay, so what we'll do is first look at basically what we say of creation in life in mind at a

conceptual level, what is the history of the concept and what they are meant to do. Then we'll look into the

formal specifics of how we try to represent basically cognition as inference and what limits

these attempts have. And finally, we will take the reflection further and see whether we can and how far we can

represent physical creation, the construction of physical reality, with the tentative formalism we have

framed here. So let us look at creation in life and mind. Historically, the term of creation, it refers to

autopoiesis. That is, that means literally self-creation and that is an attempt by an activist, what became an

activist, so mainly Maturana and Varela, to ground the study of cognition, the study of biology and the way

well, the living system work. They framed autopoietic theory as basically the idea that cognition is

grounded in a way that the living system constructs itself, like their metabolism allow them to reconstruct

the structure. And the structure does not have to be conserved by metabolism. So you have self-creation.

And this is basically the concept that tried to articulate to ground cognitive science. And later,

Varela and colleagues tried to basically concretize this by working on how the cognition was enacted,

which is implemented in the body and the concrete activity of living systems. And this basically

was very determined in the history of cognitive science and it framed the 4E approach. So the

extension of cognition beyond the brain, let us say very, very tentatively. The idea that they frame is

that you have, you can see the diagram that is taken from DiPaolo's work and I tried to explain the relation

between self-constitution. So let us say it is a weaker, lighter, more specified notion of autopoiesis

and agency. The idea is that you have self-constitution, that is the fact that the living system

builds themselves and it is basically afforded by the way living systems coupled with the environment. And

agencies, later, higher order of regulation of this disability to define basically the terms of your

coupling with the environment. And the idea it generates is that living systems, there are systems that

have a structural identity that they actively generate and sustain under precarious circumstances. There is no

benevolent universe that helps life, that helps life be life. Life has basically to generate environment

where it is life. And in doing so it generates and it sustains what it is, its identity. And all of this

is necessary for sense making, so let us say cognition, to be something, to occur.

And this is, in my opinion, to some extent in DiPaolo's own opinion, somewhat wordy. It is a lot of concepts,

a lot of Greek words, and people have tried to frame this in a way that is more mathematical and formal.

And the question it generates is how to formalize this self-creative property of biological organization and cognition, and in a way that is basically grounded in a material information flow, in and in a way that is more complex and complex and complex and complex and complex.

We have a pretty good answer to that, actually. We have a pretty good work that made a point that you had related this somewhat high-level concept to basic physical processes. This work is the work of Moreno, Mossio and Monteville, mainly, who work in the institute of Ensoli in the Pays Basque in Spain.

The idea is that living systems are constituted by a closure of constraints. So, the constraint is a physical thing that shapes how other physical things work. Very simple. If I have air in a box that is closed, the air can't leave the box, it is a constraint. Okay. And the property that individuates living things is the fact that they are constituted by a set of constraints that canalize metabolism in a way that reconstructs the same system of constraints. And so, from there, you have self-constitution, you have also structural, let us say, notion of biology that does not depend on the matter it is made of, but on the way it is sustained. That is, I think, pretty good. If you want to have a physicalist notion of life. And we have a very basic ontology in which constraints, they shape metabolism and they shape information flow, as in the way living systems react to things. And that grounds the autonomy of living things, the fact that they can recreate themselves, basically. But nothing gives a clear link between this and cognitive meaning or anything we could recognize as cognition. So, this is what we will try to articulate next. So, how do living systems, as we just defined, can create meaning?

The notion I would like to import is the notion of active inference, which I will not get into the mathematical detail here. But the idea is that the way the flow, the dynamical flow, literally, that is entailed by an agent and its environment, it individuates states that are meaningful. Basically, an agent will predict, in a relatively formal sense, sensations that it will...

I don't know how to say that, not perceive, perceive, but undergo, and actions that it will undertake. And these predictions are basically what underlie meaning for the agent. It underlies what states will just so happen to shape its own action and be meaningful to them.

This can be framed by, in a pretty simple poetic move, by stating that biological and collective systems, they do not simply infer what is, they bring about their own reality, by actively inferring what is meaningful to them, by actively looking for states that they think are really than fault, their own structure. And then you start to have a stronger relation between what a physical system of constraint is, and how it individuates states that are meaningful.

So, you still don't have a formal theory of basically how this metabolic flow builds learning or creation, like builds a new space of possible states that is not the one it began with.

So, I have a description of a structure that is the set of constraints, and this produces basically an activity of the system, a flow over the structure. And this flow, it somehow constrains or shape the structure the system takes at time t plus one, which itself produces a new function, a new flow, etc. etc. etc. etc. So, it is this dynamic of, I don't want to call that self-refragment, unfolding, whatever, the way a system of constraints produce a flow that reproduce a system of constraint that we want to formalize. And the question we could reframe as such,

we could reframe as how does cognition, the ability to understand, integrates information about the world it enacts within biological organization. Because cognition is basically the flow of biological organization. At least it is not detailed by it. Everything that I see is something that I, let us say, bring about, that I look at because of the biological constraints I am. And therefore, the question of creation can be reframed as how the meaning I project onto the world is brought back to shape my own structure. And this is what we will try to frame mathematically in the framework of Active Imprints. Daniel, any questions?

Daniel, any questions?

So, the attempt I would make here is to see if we can, how basically we can represent creation as a process of inference,

which is likely the most minimal, highly recognizable way I could frame cognition

as something that is distinct from simply its material implementation.

Let us look at a piece of math that is called the Frenzy principle,

and that is the formal motivation or grounding for active inference.

It will be the hard part for people who do not do math,

or people who are not familiar with the Frenzy principle.

I will work with it slowly.

It is a piece of math that frames statistical inference from the Mekhol system theory.

Mekhol system theory is basically the idea that you have states and they move as a function of each other.

And what the FEP does is to prove that if I have something that is called a Markov blanket,

so a boundary that any information must travel through for two subsystems to communicate,

practically, then you have a synchronization that emerges between states that are outside this boundary and states that are inside this boundary.

And not only this, but if I measure the basically,

every state that is on the inside of the boundary entails belief,

entails a single distribution over the external states.

So, the flow of internal states is such that it minimizes the international distance between the actual external states.

Well, what I can observe and what I believe to be the case,

where I, of course, is the internal systems.

So, that was abstract.

The basic and formal meaning of this is that if I have a Mekhol system and I have a boundary that acts to mediate information exchange between two subparts, then both subparts can basically represent each other and hold belief in each other in a way that is constructed by the systematic synchronization across the boundary.

It is quite abstract and quite minimal, but it is a foul proof that we can have representation,

we can have cognitive meaning from a very, very minimal set of hypothesis, which is basically a consistent causal structure.

And it is a pretty good case that a basic system, for example,

this type of constraints, constraint will be more general, but this kind of very basic non-cognitive things can entail cognition.

It has been argued that it does underlie agency and creativity and similar concepts that would entail, in my opinion, creation as a prerequisite.

So, we'll try to see whether the FEP actually can represent creation as the ability for agents to individuate states that are of meaning to them.

The thing is that the existence of Markov-Blanket is demonstrated for any complex stochastic chemical systems, so chemical systems that are random.

So, this is a very general formulation.

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It is a very general formulation.

Arguably, anything that exists can be framed as such.

So, we should be optimistic for the ability of this formalism to represent the cognitive the production of cognitive meaning by basic biological system and therefore creation.

It is not the case because when I frame something as x of t equal something something, well, I have to pre-define x .

We have to define space of states that is x .

I have to define creation as the ability to construct physical possibilities.

So, by construction, if you frame something within the Michael system theory, you have to indicate states a priori.

What you have written down is not creation.

It can be a ground for creation.

It can be interesting in the study of creation.

It is not creation.

It is not the construction of physical spaces.

And it is likely not the construction of cognitive meaning, at least in the sense that an activist and autopoietic theory people mean it.

It just underwrites what becomes measurable for a given subsystem within the stochastic chemical system, which is a lot, but is not, again, creation.

Let us think about how we should go, we should think of cognitive creation.

There will be more math here, I'm afraid.

But if a system is to observe another system, it will have to lose information about its broader environment.

So, it is a pretty easy, well, it is something that can derive from first principle.

Because, basically, if you frame it as a quantum system, observation will entail entanglement, by definition.

I observe a system and it becomes something that has individual states because I become entangled with it.

So, its states become basically synchronized with my states.

But I cannot synchronize with everything all the time.

So, if I act so as to have more knowledge about the object, I act so as to have less knowledge about my

environment.

So, there is no, basically, lossless observation.

It is something that can be derived in the quantum setting that can be reframed in other way, in classical settings.

So, if I act so as to have more knowledge about the information, because of the way observation will either induce heating, information loss,

or cause myself to forget information, you can pretty much derive it in a variety of cases.

It is likely true in all cases.

So, it means that observation, as it is constrained by my own flow and the way I couple with the environment, it does not simply entails the observation of pre-existing, well-individuated physical states.

It brings about well-individuated, well-defined observables for me to observe.

So, the way, how to say, given this, it is hard, maybe not possible, to frame a theory of observation or measurement of cognitive meaning, it will not be general.

And I pretty much have to work creation on the very fundamental scales at which my theory operates.

So, I will use the same thing.

And this is something that is arguably done by reframing of the FNJ principle for generic quantum systems by Manley Crissfield.

The idea is to use category theoretic equivalence between lots of things.

I will definitely not get into detail of that.

To reframe the synchronization dynamics that builds cognitive meaning according to the FNJ principle, as a series of binary measurements are equivalently symmetry breaking operators.

So, the way a Dymical system synchronizes with its new elements under the FEP is reframed here, as basically a series of binary questions that I ask my environment, and that enforce my interaction with it, in such a way that the answers to my questions become physical facts.

Which you can derive from a rather general quantum information theory background.

So, I will say that the FNJ principle is a very important thing.

But that gives basically no structure whatsoever to the measurements.

Nothing in this theory says what kind of system can ask what kind of questions.

And this means that, okay, this drives creation maybe, drives individuation of physical possibilities, physical observables.

So, it does not constrain at all what this creation is, so it does not explain much, and it does not help much to understand how a living system recreates itself.

So, we still have somehow to recruit the cognitive and demical formulation of the FEP that I discussed very tentatively earlier.

So, let us take things from there.

I have turned, during this formal discussion, from the notion that we have to represent creation in the living and cognitive world,

to the notion that somehow it is necessary to represent creation of physical reality, like the underlying states of physics.

So, let us discuss this.

So, let us discuss this.

This third part will serve mainly to discuss this claim and what it means and what we should do with it, basically.

First, I want to show basically that the claim that there is no a priori physical reality is a pretty basic claim.

It should not be treated as exceptional. It is necessary property of a coherent physics.

It is necessary property, actually, of the formulation or description of a world that is beyond what we call metaphysics.

Let us be more specific in the argumentation.

Let us say that the argument is not a naturalistic argument.

I call this argument from ontological consistency.

An ontology that is naturalistic, that serves to present nature, if it is consistent, cannot call onto external objects,

objects that would not be natural, to explain the properties of its own objects.

For example, if I tell you this thing is black because it reflects the light in such and such way, and I can also explain the properties of light, it is self-consistent.

If I tell you this thing is black because God willed it, and I cannot explain you what God is physically, it is not self-consistent.

There is no property that is framed by Alexey Rosansky, who is present here as the property of self-function consistency,

that is that the ontology should basically include itself.

The discourse we have on physics is, unless we have very radical position about the relation between human and the other world,

it is something that exists in nature, and that is something that we need to account as an element of nature.

This entails two things.

This is one thing, actually.

It entails that scientific representations, the models and theory we have, and the property those models and theory have,

they must be accounted for as natural entities in their own rights.

And they must be, at least in principle, explainable by the tools of physics and science.

And this entails, in turn, that I cannot have such a thing as a law, a symmetry or a set space that pre-exists physics.

Any statement I make about physics must itself be explained by physical processes.

I will elaborate on what it means.

The view of physicists is typically that we have theories, and those theories are representations of the world, that just so happen to account for fundamental structure of the world.

For example, we have the Maxwell law of electromagnetics, electromagnetism,

and those laws are representations of a force that exists, and that is generated by SU1, I think,

by a symmetry in the Hamiltonian, sorry, a symmetry in the quantum property of systems.

But this symmetry pre-exists basically any atom to represent and any physical process.

It is a fundamental property.

What I say is that you cannot have a fundamental property.

You cannot have something that predates physics and that cannot ultimately be explained.

And if you admit such a thing, your ontology is not consistent because you have to admit some things that are beyond that predate nature to explain things in nature,

which is the opposite of what self-consistency is.

So, if I want to frame a world where you do not have explanations that call onto things that are beyond physics,

you have to have law of symmetries and spaces that are constructed by physical processes.

And it is not a problematic view to have.

At least epistemologically, it is necessary.

Let us discuss what it means concretely.

Let us discuss what we have in the context of the system.

We have cognitive systems.

Those cognitive systems, they perceive.

And this perception brings about cognitively relevant states,

which the cognitive agent experiences as real.

We have physical observers.

These physical observers bring about what we call pointer states,

that are well-individuated states that are measurable.

And those states constitute physical reality,

at least as any given observer can experience it.

What I say here is that this is not epistemic notions.

The punditent states and the cognitively relevant states that perception,

the cognitively relevant states under perception by agents,

that are real and they are individuated as real by their observation.

And if we frame it properly, the QFEP, at least what we could do with the QFEP,

it represents how protominds, at least particles, agents that have properties that we classically

understand as mindful, they individuate physical observable, which means that they create spaces,

which means that this is a theory of creation, not simply a theory of how access reality,

but a theory of how we get reality in the first place.

So, I will try to do anything like grounding properly this claim.

It will take a lot more details.

And I will try to work by showing what the searcher theory would do,

because obviously I do not have such a theory that is framed formally from start to finish as of now.

And if I did, I will not present it now.

Let us look at the way constraints basically create the relative experience.

This is a diagram of how us humans in the world, the social world.

Basically, we understand things like constraints over what we do and can do and what it entails.

I understand that if I talk to you in proper English, you will understand.

If I talk to you in French or Swali, you will not understand.

Understand that if I undress, it will be inappropriate.

And if I hold myself in a certain way, it will lead you to believe me more or less.

Understand that if I start speaking very loud, it will disturb people and they will act for me to stop talking very loud.

I understand very basic facts about how this world works.

And I understand how this constrains the action I can take and the consequences of that action.

What I experience is basically a social cultural landscape, which exists independently from me and that I act within.

What I claim is that this social cultural landscape evolves in a way that is creative, that is basically obvious.

New norms exist that did not exist 10 years ago and that is even more true for one millennium ago or before human were a thing.

New norms exist that are not just a social cultural landscape.

And the way I can represent how I try to understand this social landscape and possibly fail or erode norms and the landscape in such a way.

It is equivalent representation of the landscape itself and the constraints that constitute this landscape.

So what I did here is to basically frame a theory of creative evolution, this of a set of constraints, well of a social cultural landscape, on the way individual agents act, understand and act within this landscape.

I have a sense in which constant observation and cognition about physical, here a social reality, but still physical in the sense that it exists in a physical world.

How this landscape can be created by something that is cognitive and this basic ontology, if I am correct here, is something we can

to do and mathematize and that we can frame as a basic theory of the individuation of new physical possibilities.

Credit to this graph is given to Ana Brinac.

And what is of importance here is the duality between two representations, two perspectives.

the perspective of agents that basically enact social-cultural constraints,

that they experience as a physical fact, as a reality,

they cannot go beyond or escape, or maybe affect in brute force,

but not escape, and the perspective of a set of constraints

that intentionally unfolds in a way that is creative.

That is a formal intuition I plan to build on.

I can say something a bit more specific about how it works

by analogy to the FNJ principle and the holographic principle.

Basically, creation operator, in my opinion,

I could argue for that, I will not do this in detail right now,
is something that goes for a space of...

It's not actually a space of physical possibilities,
it's a space of potentialities.

It does not have possible states and impossible states.

It is just the statement that something is possible.

It is very close to what Greeks called chaos.

So you have such a space of...

...unindividuality possibilities, and from this space unfolds a pair of an individual observable,
which is called E for environment, and an agent that is capable to observe this observable.

And the agent is also capable to maintain a boundary between itself and the observable,
or basically the observable to stay existing as observable, to be observed.

And from this, I think that some mathematical operator that has this basic flow is capable
of the construction of physical state spaces as they are unfolding strictly,
strictly, no, metaphorically.

A physical structure as basically the...

Yes, the emergence, the co-construction, the individuation, as you will,
of a well-individuated observable state, an observer that is capable to see this state
and bias in this state, maintaining whatever worked as an individuated space of possibilities before
as a well-individuated boundary between itself and the system.

Let us think, for example, of a piece of paper.

I, an agent, can basically write things on it.

And by writing things on it, I create a new physical observable,
which is the content of what I've written.

And the fact I write on it and or read it makes a piece of paper,
not a piece of paper, but a boundary between myself and a semiotic space
where the writing occurs.

This is the basic dynamic that must be framed for us to understand cognitive meaning
and that, if we framed in such way, could work to describe
the construction of state spaces ex nihilo.

Because, literally, you do not need physical possibilities to be a thing or be to exist.

It is the point of this presentation.

You don't need a space of possibilities to pre-exist the individuation of the agent and environment.

This is why it is relevant and this is why it is hard to formulate.

And given an operator with this property, we can pretty much work it back
at the foundation of physics and see what happens.

So, how about physics of creation?

Let us tie the threads of this presentation.

I will formulate three take-home messages.

First, we have something that is called the FEP, that is now framed, that is mainly framed as Bayesian mechanics, as informational enrichment of Dimi-Cole System Theory. But that can also be framed as something in quantum information theory.

It says that agents actively infer states of meaning through their engagement with the environment, in a way that maintains the structure.

This gives us a formal background.

The quantum framing of it gives us a very, very tentative and, how to say,

I will not say ill-specified, but very few specified.

It does not have a lot of detail.

So, it gives us a very not detailed background for the study of how cognitive meaning underwrites the construction of physical set spaces.

But because it is very abstract and not very specified,

we have to relate it in a much stronger way to the classical formulation of the FEP,

where the internal flow of the agent and the synchronization manifolds that emerge from the interaction, have a very specific topology, have a very specific geometry, and can be equipped with meaning about the underlying physical structure.

Second, we have to admit that past-patri-realism is a thing.

We have to admit that there is something wrong in how we view science.

Let us quote Einstein.

If, without in any way distributing the system, we can predict with certainty

the value of a physical quantity, then there exists an element of reality corresponding to that quantity.

By this account, nothing is real, which is a very big problem, a very bad position to hold.

And we have, in some sense, to account for the role of observation in creating, well-invibrating physical states.

Not only for quantum physics, not only for cognitive science, but for everything in between,

which is, in some sense, entailed by quantum physics, in some sense entailed by cognitive science.

So I don't think it is a very big deal to say that, but I think it will be taken as such.

And the statement I want to make is that not only doing away with this brand of realism is compatible with naturalistic demarch,

with science, but it's also necessary.

Because for naturalistic science to be a thing, I have to represent myself, representing as there is something metaphysical that comes into play.

Either as taking myself as a detached, non-physical describer of the world, or by postulating some kind of godlike pre-existing structure,

that I just so happened to stumble upon, you will have something that is beyond physics in some sense, if you don't do away with physical, with this brand of scientific realism.

And it has been said to me that it is not a new view. No, it is not.

It has been framed as a criticism of the notion of Atman thinking itself by an early Buddhist philosopher about three millennia ago.

And the fact it is not new is not really a problem, in my opinion.

We should go with things that seem true, whether or not they are exciting and novel.

And the fact that we can have a cosmology without a presence in nature, and other people do not find any problem with it,

should be a sign that maybe our insistence that no, there are things in themselves, is not well motivated.

And finally, what I propose here, beyond philosophical considerations and discussion of what some piece of math do,

is a very, very, very, very basic, not draft, but drawing of physics of creation, that holds in two movements.

The first is abstractly the individuation of something from nothing, of order of chaos.

Given a space of undefined possibilities, B, you have the individuation of an observable that is well defined, and of an agent whose observation of E also individuates a boundary.

So, we have this kind of framing that could work as a handle on how to write suspense construction.

It does have a very strong category theoretical filler from the way, basically, the content of B is not what's at stake.

This diagram is pretty much agnostic concerning what B is and is not, and it must be because it underwrites the becoming of B.

And there is a lot of work, if I want the physics of creation to be a thing, in framing properly what kind of mathematical operator

basically follow the flow of this diagram, and what those mathematical operators do, and the kind of physics they can underlie.

But for this, I have a concerning case. I have the case of how humans collaborate to build and recreate social norms,

because it is a case of creation, like it is pretty much the single case of creation I could convince anyone is a thing.

There is money because we believe there is money, and it works as we believe it works because we believe it works this way.

It is a very trivial statement to say that norms constrain us because we believe in norms.

And the duality it affords between the representation of single agents trying to understand and act within a social world,

and the flow, the interesting flow that is created by social constraints and that recreates social constraints, it is pretty specific, it is much more specific than the vague operator on the left.

And I know this operator must represent that. So it gives basically some level of grip, at least an entity level, on how the physics of creation would work. And they would work by describing the unfolding of physical structure

as the construction within physics of meaning and or constraints over what is observable by given agentive observers.

And pretty much we have to work from there because I do not think there is no self-consistent in the sense I introduced earlier.

There is no self-consistent theory of physics if we don't do that. So let's do that.

This work again is to some extent my work, to some extent the collective work of Kairos, which is basically, it is a perspective laboratory that works towards an active sense, an active naturalism. A sense that is naturalist in the sense that it plays the observer within the world and works within well-defined causal ontology, rather than less well-defined laws and abstract principles.

And it tries to do so in a way that enables people, as in humans, to understand their reality, to basically use this knowledge about how the world works to constrain their own behavior and navigate the world.

Especially given, well, the recombination of constraints that will, we expect, be at play in the coming decades. So I strongly suggest you take a look to our website and our general line of work and consider if you want to talk to us or join us.

The QR code is, I believe, a link to our website.

So thanks to Maxwell, thanks to Daniel, thanks to Alexei, thanks to Yona, thanks to pretty much everyone who helped me write this and think it through.

It was big work.

So I will run through this slide if someone wants to read them in the video.

And now we can go toward the discussion.

Excellent. Thank you.

Thank you.

Thank you.

Alexei and Maxwell, welcome back if you want to join.

And please, either of you, provide a first statement or any set of questions, however you want to continue. Just go for it.

You know what? Give me five minutes. I will try to destroy my head. So chat, but don't expect me to.

Okay. Well, do you want to share your perspective on the project? Things that Aval would be familiar with while you're here?

Yeah, I can begin, I guess.

Yeah, so I worked on the metaphysical or ontological parts. And yeah, to me, Aval gave it a good presentation. Yeah, it was really nice to see it framed into the whole way of talking about everything.

Like linking it to FEP and then to other stuff.

So, yeah. So the point of the, if I can share a bit more about the metaphysics of the ontology.

The idea was to create an ontology that is naturalist. And as FEP seems to be really naturalist, then I wanted like to make it work together. At first, it was only in my own field that I developed this ontology. Then Aval said to me that it was really linked together. So that's what we tried to do here.

Yeah.

What does that look like? How are you evaluating what a metaphysic is or how you know when you found the right one or an incompatible one?

Yeah. So the basic question was about the question of what exists more generally. So that was my, my anger

of attack. If it's a thing in English. Yeah. So there are many discussions about what there is in philosophy.

So I think it's a bit, how to say that, mainly in the 20th century, mainly in the 20th century, due to crime. I think it's, is popular in general in the culture of science.

So I think it's a lot of people who are in the 20th century, but I think it's a bit of a lot of a lot of views about what there is and what there is not. So I was a bit, how to say that, disturbed by the fact that there are things that are not.

And I wanted to give it a try to try to talk about what seems not to be and to make it exist in some way. And so, how to say that better? Like, generally, what doesn't exist is linked to consciousness, like Pegasus, or let's say, stupid sentences, such as the round square copula of the college.

Or something else such as, I don't know, some weird, not really mathematical entities. And I was like, okay, so I admit that they do not work. But at least we are talking about them.

So, can't we just say they exist in some way? And then I went into social ontology. And I found that we can call them social artifacts. And then I was a bit relieved with this view.

So, I tried to make it work with also phenomenality or cognitive or personal experience, human experience. Because being a social artifact doesn't make it all. So, yeah. Then I just tried to make it work between ontology, social ontology and cognitive ontology.

And yeah, that's it. And also, yeah. Something else. I've talked about the self-referential inconsistency. So, it's a debate I've found by working on ontology. And I was really into it. The fact that the theory is inconsistent if it doesn't apply to itself.

And I was really into it. And I found some articles about it. And I found some articles about it. Which were against naturalism. But one kind of naturalism only. So, it was mostly eliminativism.

And I saw that the people who were against naturalism. And I saw that the people who were working on that, they didn't know about a lot of naturalism. And as I work with Kairos and in my study, I found that there are many kinds of naturalisms.

So, I saw that the argument doesn't work with other kinds of naturalisms. Such as the one that is developed here. Yeah. I hope it's clear.

Yep. I'll ask one question on this topic from the chat. And then feel free to take it in any direction or Maxwell. So, Adam Rostowski in the chat asked,

Thanks, Avil. Question maybe for Alexei. What if state space is considered theoretical ideology rather than causal ontology? Its presupposition would then not be a threat to ontological consistency.

So, what if state space is considered theoretical ideology rather than causal ontology? Maybe just unpack the question and then give it a thought.

Yeah. I will write it now in order to reflect on it.

Perhaps, Avil, just a direct question. What are the state spaces that you're concerned with? How would you characterize what they are?

So, by construction state spaces, there are spaces of postulated physical possibilities.

I do not understand the notion of ideology. How it works here.

It's pretty trivial that what we call ideology will define what a person or political system or institution is capable or interested in seeing.

It is the topic of Scott, seeing like a state.

A foundational work of anarchist anthropology that looks at basically how whenever states try to do things

overtly usually for the population, their main concern in effect is to build the capability of control.

And usually they do so by cutting through complexity aggressively.

A very basic instance of that is scientific forestry, that is.

I can say it is indeed forestry.

The idea was that Prussia and he did warboats, loads of them.

He also needed to be able to predict how many warboats they would build on a given timeframe.

So they need to know how many warboats compatible trees they had.

But you don't have warboats compatible trees and not warboats compatible trees.

You have trees of different size, different ages, different essence.

So if you just counted trees, even with a well-defined measure, you would not really know how many trees you have.

So they just raised forest and planted a much of a patch of a given species at a given time, with a given soil.

So they knew that when they knew they had 64, I don't know what trees there are.

I actually know what trees there are. I don't know. It's told in English.

So when they had, it's not oak, but they will say oak to say something.

So when they knew they had 64, 30 years oak, well, that's 64, 30, I did say 30, 30 years oak.

And that's it. They can cut the trees and they know exactly what they can do with it.

So this is a way in which what something we could account as ideology.

So the notion that we should be able to control blah, blah, blah, blah.

So the trees basically construct physical reality in very concrete sense, because they cut trees, they plant other trees that fit their expectation.

It is a form of niche construction. It is a form of cognitive accession in the sense that the loads, they would need to account for how trees are they just offloaded onto the environment by making trees actually

all the same. And it is a form of meaning projection.

They have a system of meaning, which is that they need to count trees that are barbought compatible, and they project it onto a bike. How actual forests are organized.

So those are physical possibilities as an interest by the state. Those are state spaces.

We have a state space that was very complex with a lot of possible trees, possible species, possible size.

That is brought down to how many trees I have. And it is...

Could you repeat the question word for word, please? Because it was not clear to me how the national ideology intervenes.

What if state space is considered theoretical ideology rather than causal ontology?

Its presupposition would then not be a threat to ontological consistency.

The causal... Yes. Okay. I understand. The causal ontology precedes a state space.

It is the ontology of creation slash constraints that I have waved at somewhat imprecisely at the moment.

State space are, I think, I think I will be able to demonstrate it, but I can't at the moment.

They express the point of view of an agent. So I imagine that this is what is meant to be

introduced by a ideological theory, theoretical ideology.

And I do not think state space are a threat to ontology. I think ontologies that consider state spaces as fundamental are flawed.

Ontologies, fine.

Thank you, Avel. Alexei, feel free to chime in. And then Maxwell, if you want to add anything.

Yeah. So I think I will join Avel on the commentary.

Also, I would like to say that if we take Quine's definition of ideology, if I will not explain this now that...

Well, yeah. So the idea, I think, is that the concepts are a tool and not fundamental stuff.

That's what I think it is implied by ideology here.

So what I'd say is that you give the value of being an ideology or consul ontology, depending of your goal and your criteria.

But at least minimally, the stuff you're talking about, it is. That's the fundamental stuff that I would say.

Yeah. So even though it is ideological, even though it is instrumental, let's say, minimally it is something and it has consequences on the world.

And that's what I think Avel said with the niche construction of social time.

It is... It didn't really say that in this world. But yeah, even though your ideas are, let's say, stupid or absurd and people appear to hear you, then your ideas, they are in the world and they do something.

Yeah. So I hope I answered well. If not, then I can still take questions.

Thank you.

Maxwell, do you want to add something or...?

I don't currently have anything burning to say. Avel, I think this is the...

Well, this is the nth iteration of our discussions around this.

I mean, as always, I think you did a great job presenting what I think is, you know, a genuine issue.

And I mean, I've just been rehearsing the same, you know, appeals to multi-scalarity.

I mean, I find it interesting that you would speak an answer to these issues in quantum FEP, precisely because of its scale-free nature.

So, you know, you can't appeal to a space-time background when speaking in the quantum FEP formulation.

So I think that, in my mind, that makes things potentially more difficult.

I see why you would want to do that because of the focus on, you know, the definition of an observation as a physical interaction.

I guess, um, my, uh, my kind of, uh, overarching question is whether the multi-scalarity of classical FEP helps at all.

Uh, in the sense that we've described, uh, before. So you have a set of observables.

Some of them change too fast to really like disclose or allow us to track reliably some phenomenon behind the blanket.

Others do.

Uh, in the latter case, we talk about states in the former, we talk about random fluctuations.

Um, so I suspect that, you know, the, the kind of definitional, uh, discussions like, uh, by which I mean, like, you know, definitions of states, like these kinds of foundations.

These kinds of foundational issues that arise, uh, in classical FEP might be helpful hereabouts.

Uh, I also suspect that you'll disagree with me. So I'm looking forward to hearing your thoughts.

So, um, basically, uh, you, you have, uh, the distinction is clear between us.

At least as you frame the problem, you seem to be a realist about, uh, physical possibilities about states spaces, which I'm not.

I have an argument, uh, for non-realism, which is, uh, self-frontal, well, ontological consistency and the two arguments of self-consistency and self-frontal consistency.

That brings us to say that you can't have preemptive states spaces in which, uh, to frame, in which to classify, have physical observables that somehow pre-exist physics, nature, the processes that are within it.

So that is one, uh, physical arguments. The question is, uh, then can I demonstrate conclusively that you do not have physical observables before observation, but state spaces do in fact, try to use the perspective of an observer.

And basically if I can demonstrate it, uh, the, uh, debate is over.

Um, until then you have, you can basically, uh, hold, you can hold structural realism, but you have to account, uh, of how things exist.

If they are not fundamental, I think, uh, multi-scale FP can do that. So it's not a critical problem for you.

Everything holds on the, uh, formal demonstration that we do not in fact have, uh, set spaces as a primitive physical effect.

Well, uh, just, I would push a little bit back against your realism versus anti-realism, uh, distinction here.

I'm really just talking about, uh, well, I'm, I'm really just talking about like at the level of the formalism, you start with some of the observables.

Um, and then you, uh, can talk about the way that those observables are related to, uh, some other things, which may be latent.

Um, all of that is a formal statement as far as I'm concerned.

Um, and like, it is useful whether or not it maps in some kind of fundamental realist ontology sense.

I guess like if, if there is a, there is a kind of transcendental overtone to what I'm saying in the sense that I, I, you know, we've argued in print in this on the map territory fallacy fallacy paper that the, uh, the FEP theoretic maths and, you know, the, the related cluster of principles, unitarity, maximum entropy, and so forth, provide alternative.

Constraints on any model building exercise, just full stop.

So again, that, that, that doesn't presuppose any realism about what maps map onto.

It's really just a statement about like the construction and maybe the process of constructing.

Um, you're correct.

I wasn't, I was, uh, uh, uh, attributing.

I don't know what to say it's English.

It was not accurate to frame you as a realist about spaces.

It was correct.

However, to frame you as an instrumentalist, which I refuse to be, uh, because of naturalism, basically, I consider it necessary to frame causal ontologies that map onto reality properly.

As fine-grainedly as we can.

And to do that, we have to include basically the translation relation.

And this is when everything becomes, uh, complicated because you can't say, uh, if there are such physical possibilities, then this occurs.

You can't use that language because by doing that, you frame physical possibilities as something that is abstracted from the underlying, uh, physical reality.

You, um, decouple the formal from the actual.

And I think whatever ontology we use must be what reality is, must correspond structurally to what drives reality.

So I have, uh, it is the contrary.

It is, uh, I, who is the realist, but the constraint.

I was going to suggest, I mean, maybe, uh, maybe actually the, so I guess, uh, you know, basically.

This clarifies things a bit.

I go for an ontology that is consistent in the sense I find and the constraint over that are much, much, much, much, much stronger than the constraint over an instrumentalist ontology that just, yeah, we say interesting things, which is not actually easy.

It's just much more easy.

But suppose we started from the assumption that we don't have to have a strong position on the reality of state spaces.

Uh, does that affect your argument at all?

Uh, so, you know, if I were to say like, I, I just, as a practitioner of science, start from my observables.

You can be a radical phenomenologist from this point of view, if you want, you're just starting from your observables and then working out like what resists your manipulations or interactions.

Um, yeah.

Does that in your view kind of allow us to move around asking the questions about the physics of creation?

You can move around with, oh, about the physics of creation?

No.

Um, move around.

Yes.

Understand some things with cognitive meaning.

Yes.

You have a FEP grounds cognitive meaning in the micro-system theory.

It's a huge deal.

Um, what it, um, what it does not afford the ontological consistent, uh, theory of physics.

And the difference between us is not what we hold the FEP to do.

I think we're both instrumentalists.

Um, you have a FEP grounds from you as a realist.

Um, but the difference is that I see an affordance to build a much stronger, uh, self-consistent, uh, physical theory that draws elements and the general philosophy from the FEP, the philosophy of active inference,

basically.

And so what you need here maybe is, uh, excuse me.

I push in this direction because I think the goal is variable.

It is ultimately an ethical call.

If you do not agree that the goal is variable, you should not agree that it is useful to push it.

I do not have ultimately an issue with that.

So, um, maybe, um, there is a kind of Kantian take, um, here.

Cause I, I, I, I mean, I think in the, at least in the history of philosophy, this, uh, split between realism and anti-realism or instrumentalism.

I mean, uh, like, I think it's perhaps overly simplistic as a description of the, uh, potential options.

Uh, there, you know, there is a whole Kantian line of thinking, uh, the, the talks about, um, you know, something intermediate in some sense.

Um, so, you know, yeah, I don't know.

Uh, just a thought.

So, um, I couldn't elaborate.

Is, uh, Adam in the room?

Adam, if you're in the room, uh, use the chat.

But in a sense, I agree that.

Not in the room, but I can read another question whenever you're done.

I'll read another question from the chat.

Yeah, but I was going to say something.

Which is that, uh, I agree that the, basically the distinction between, um, instrumentalism and, and realism must be dissolved because both, none of them are ontologically consistent.

Both suppose a representation relation that is outside the world.

And the question is whether the words as a formal construct correspond to the reality as a structural thing.

Um, it is not a constant question to ask the, all of the things are in the world.

We have to have a theory of how things are true or not that are false for the statements we have about the world to be in the world.

Because they are.

Um, and so from the moment to aware, from the moment you agree that, uh, ontological consistency is something to have, a lot of the distinction we have in, uh, Western philosophy will be dissolved.

Which makes communication complicated, of course.

Maxwell, you want a closing thought on that, or I'll ask the next question.

Um, you can ask the next question.

I think that, that, that brings us, uh, nicely to, uh, yeah.

This is from Alex Kiefer.

So Alex wrote, I love the positive story in this paper.

Insofar as I understand it, which I'm still working on.

And the idea of treating even such things as state spaces and scientific reps as amenable to physical

descriptions.

But to bounce off what Maxwell is saying, I'm not sure I see the deep critique of thing in itself realism here.

Probably depends what one means by that.

I tend to think of the strictly incoherent Kantian idea of something unknowable behind phenomena.

So why is treating state space creation as part of phenomena at odds with that?

And it's also pasted in the chat.

That's, uh, just to pick up on that, I think, um, you know, we, you don't need to be a strong realist.

Uh, you can just believe that, you know, you are more or less getting it right.

And, uh, the semantics that we've proposed, uh, you know, that uses active inference and the free energy principle as a kind of, uh, formal basis, but precisely uses the fact that, you know, you have a certain perspective, you have certain beliefs and your beliefs are going to be more or less wrong.

Uh, and the, the, the prediction error or the free energy is precisely what, like what you need to, uh, the kind of minimal ingredient that you need to hook you into a situation enough to, uh, you know, be doing something.

You know, pragmatic that you might call truth telling.

Um, I think the, uh, yeah, that that's really all you need from a formal point of view is just the ability to be wrong, um, and to change your mind in light of new evidence.

So my, my claim is not that everything in, in physics is false and, um, formal statement about postulated possibilities of no epistemic value or trivial epistemic value.

My claim is that those representation as any representation are, um, rational, they relate to objects.

And the truth value or value in a sense that we have to give the meaning we have to give to that, uh, if we are to give anything at all, it must be grounded in the study of physical dynamics that underlie the emergence.

If we are to have, again, an ontology consistent theory, we could just say that science is discourse that is not entirely false and say things that are, uh, not entirely false.

I could do that.

Uh, again, the notion that series, uh, physical discourse must be ontologically consistent is ultimately ethical.

Like it is a statement on what you should and should not do.

I can say that it is better because X, Y, Z, you can say that X, Y, Z are not meaningful targets to have.

And that whatever we do is okay, essentially, or maybe it's better than X, Y, Z.

Uh, but the notion that I am, um, I do not critique state spaces, but at least it's not the point.

I have to frame it as a critic to, for it to resonate.

Uh, but the point is that state spaces are rational property.

And that to study this, uh, rationality, it reframes the entirety of, um, physics, it's epistemology, it's ontology, it's cosmology.

And if you look at the flow of this theory, it's not pure abstraction.

It analyze social evolution.

Social evolution is agent-projected meaning as norms.

It is something that is very basic, uh, for us to understand how the system evolves.

And yet there is no such theory.

There is no theory of social evolution that do not presupposes variation and abstract away any form of structure and metabolism.

It simply does not exist.

So the non-existence of a naturalistic theory of social evolution is maybe a pretty big problem for a purely predictive, uh, pragmatic value, uh, standpoint.

From a more abstract standpoint, we don't have a Korean cosmology.

We don't have an explanation for why, why the things and why the things that are, look like what they are.

Physically, we just have to postulate that states by, uh, space time is such and such for physics to make sense.

It is a problem.

We can't have a physical theory that is, that stands only on the fact that its statement does not give rise to obvious problems.

We have to have something beyond the statement.

We have to have some way to account for how this statement became true.

And it is a pretty basic point to make in psychology, biology, sociology.

We can't just say things are such and such.

We have to say how they became such and such and why they became such and such.

It is a nothing in evolution, nothing biology makes like nothing in biology makes sense except in light of evolution.

And it is because those are systems that are, I don't like technological, let's say teleonomical.

Those are systems that are organized, that are coherent, that work in a specific way.

And so we have to, cannot just say they do X, we have to say how and why they do X.

And this epistemology has not, has not permeated physics basically because of the belief of physicists in a platonic truth of symmetries or formalism or symmetric.

I have a question for you, Abel.

So I'm thinking about this in terms of the direction of travel of our research on the free energy principle, which I mean, there are two main components to that we could discuss that are relevant here.

So, the first is a move from state based to path based or path integral formulations where what's at stake is precisely probability density functions, not over individual states, but over trajectories, entire trajectories of the system.

So, the second is the way that we can see the path of the system, where like what your generative model or Lagrangian is describing is the probability of some specific history.

So, this kind of intrinsically historical formalization.

And on the other hand, as a kind of second main direction of travel, really formalizing these kind of three nested scales of inference, where you have inference over state factors and individual states, like, you know, color, and red, green, blue, and so on.

Inference over the parameters of the model, the characterize the relations between states and configurations of states.

And finally, the structure of the model, which is really like the the actual, you know, relations and functionality of like what is connected to what.

So, one of the things that we're, you know, working on is getting to the mathematical foundations of these basically mutually constraining scales of inference.

So, suppose, you know, you had access to a path based description of the free energy, you know, formalism supplemented by a clear, you know, a clear formulation of inference nested within parameter optimization nested within structure optimization, you know, in other words, like assuming that we solve structure learning. Structure learning, which is a big assuming.

Which is a big assuming.

But, you know, if you granted those two things, would there be still much of a problem you think from a formal point of view, if you could just like move in the space of all possible models, structures, and paths through such spaces.

Would you not basically address what you have in mind here.

Yeah.

So, three things first.

Path integral, it solves the issue of presets meaning or logical systems.

And the issue of, well, you solved that already.

But it is not something I called on here.

So, it would not help with the issue I've discussed.

Second is structure learning.

The question is how basically living things project structure.

If we have that, we would progress in a big way.

Or a theory of cognitive meaning and agency.

But you will still operate within a given space of possibilities.

You will have attraction slash contraction of the state of possibles.

By virtue of what the FEP is, it builds a manifold.

So, the manifold, like, nothing will evolve on the manifold where it will evolve.

So, some of the state space will become trivial.

Some of the dimension will become trivial.

But, and some things that would not, that do not really appear per se in the state space will become relevant.

So, they will become part of the state space as experienced by the agent.

But the hard part is to account for how this state space as experienced by the agent becomes this state space per se.

How you have a matrix of six par six that serves as a boundary, let us say.

And how agents will basically collaboratively build systems of meaning where they have alphabet and can't communicate.

long sentences using this grid.

And this is not in and of itself a question of structural learning.

Although solving structural learning was enabled to have a lot of computational experiments that would help greatly with looking at the fine-grained structure.

The question is how you use mathematics in a way that describes creation, that describes the construction of

state spaces by the activities that occur within the state space.

That's what I'm suggesting when, you know, what I'm talking about is if you could figure out when to add a new node or edge to your generative model.

Like precisely, when is it appropriate to add a new parameter?

When is it appropriate to say, oh, these two parameters are connected to that space in these proportions?

If you had that worked out, I could see a kind of straightforward resolution to what you're talking about.

I'm saying like, you know, there is the system just unfolds dynamically.

And occasionally, you know, it's, it learns new structure or parameters, which basically means like, you know, new degrees of freedom are added to the system.

You are again talking at the epistemic level.

You can describe a lot of things by basically looking at the dynamics and being smart about it.

You don't even need advanced mathematics to do that.

The statement we make is different.

Let us say that you have people that agree that shells are an appropriate means of exchange for X, Y, and Z.

Then money becomes a thing.

It is a thing that exists individually because it has structural properties and these structural properties drive observable through dynamics.

And equally inevitably did not exist in one before.

The question we ask is how we account for an evolution within a space that builds another space.

And we cannot just look at what happened in the space and declare how it has changed.

It is not the problem we are posing.

The problem is how to result mathematically.

Something that occurs within a space that produces another space that is the same, but not the same.

That is structurally similar.

That inherits from what existed before, but is different in qualitative aspects because this one thing became.

So what we have to have is the theory of becoming, basically.

We can be very smart about modeling things without having a theory of becoming.

We can be very precise in empirical inference and very efficient at seeing regime changes.

That does not mean we have a self-consistent theory of becoming, creation, space construction, call it what we like.

Those are two different problems.

Although one may help the other greatly.

Both may help the other greatly.

Because the thing we are looking at is the same way looking at two very different aspects of the thing.

Max will want to add something or Alexi on that.

No, I'm going to think about that.

Thank you, Avil.

Awesome.

Yeah, from the chat on that topic.

You're welcome.

Tom wrote a similar question.

Avil, could you elaborate more on how ACTIMPH could be expanded to accommodate state space creation?

Assuming ACTIMPH is normative about free energy minimization, when should an agent expand the state space?

And Alex Kiefer also kind of added a note.

It's appropriate to add a new parameter when the model complexity accuracy trade-off says so.

More seriously, isn't free energy measurable across models or different state spaces?

So it's almost like there's this structure learning, meta cognition layer, and it almost is like, can the box be wrapped within the box?

Can it be possible to have a state space that within its state space represents itself?

And or to what extent do these physical intermediates and their associated physics and metaphysics need to come into play for different systems and at what level of generality?

Basically about this, I'd subscribe to a variant of consistent history interpretation of quantum physics, but it's not quantum physics.

It's just something that has the kind of physical properties quantum physics has.

Well, I did commit to quantum information earlier.

So, okay, let us say it is in nature quantum, which is bit trivial because yes, it has pieces in it, which is what being quantum is, but whatever.

So, this is the most specific mathematically consistent way to present it that I found.

Do you still see those slides?

Yep.

EBA.

EBA.

Bay.

And this is basically category theory because this diagram has a property, which is that following a the arrows basically change what the space is.

B begins as a neutral interface or a space of possibilities that in principle afford interaction, but it is not semantically loaded.

And the fact that you have the integration of E, the observables and A, I framed it as an agent, but it would rather be

agentive capabilities or constraints that enable the observation of E.

B.

So, this does not conserve the property of B because B becomes equipped with semantics.

So, we have theories that look at spaces without looking at what is in spaces.

So, these are categories theories, and those are the most abstract field of mathematics as far as I know.

And I do not afford semantically loaded things like free energy.

Free energy is a measure of, not only it entails states, but it also entails a measure of states.

I think the basic mathematical structure we need to represent that is much more basic than any other things, and any

FNG measure will break down much before we get there.

But what we have that is more exponentially loaded is this, the duality between the experience of the structure to landscape by agents and the engineers flow that is generated by the landscape and reconstruct the landscape.

So, the question was, what was the question?

Could you repeat it word for word, Daniel?

It was just about the normativity around model selection and whether modeling state spaces could or would require the intermediation of the physical artifact.

The normativity is there.

There. It is in this representation of the problem.

I have a small agent that represents a circular landscape and acts within it, and I can just claim, okay, there is a set space. There is a front set space. I can define free energy.

This agent minimizes free energy.

And then this is where structural learning intervenes, basically, because those agents will try to understand the world.

They do not experience the world directly.

They will infer the world in a way that is robust and competent, but they can make up new things.

They can fail to understand. They can cheat. They can erode the structure of the landscape that is collectively enacted in many ways.

And this, again, is dual to something that is the dimical. It's not dynamical, actually.

I do not know what it is. It doesn't have a word.

The unfolding of the structure of the landscape under its own flow that, again, is not dynamical because it's not what dimical sensory says.

So we've described what you're talking about in terms of niche construction.

Yeah. So in human systems, I thought, yeah, we had a, Karl and I had an interesting discussion about this a few weeks ago.

And yeah, I think what really distinguishes human systems from non-human systems is the degree to which the active inference loop is mediated by what would otherwise be internal states.

But that end up having this kind of ambiguous function of, yeah, they're environmental, but they encode the material aspects of some normative practice or something.

They store some information about what you're doing, like a like a blackboard, for instance, or these slides that we're all looking at.

And, you know, in some sense encode the the thought or aspects of the thought that you're trying to communicate.

So the. Yeah, maybe a lot of the problem here can be finessed by noting that it's not just about agents interacting with each other.

There is a kind of common store of, you know, or traces, you know, like a kind of material thing that we all have in common.

Exactly. Yeah. And.

You have a common collective inaction of a given structure landscape because humans are such that then they synchronize in a robust and simultaneously loaded way.

And that's the landscape. It does not exist.

It is not something that pre-exists the fact that agents do that agents infer properties and then enact them.

And this make it an open landscape and the normativity of the FEP affords.

Probably a computational experiments design that help us see how such landscape unfolds and look at material theory where the agent is absent, where it is landscape that produce the landscape.

And once we have that, we have a theory of creation. It is what the theory of creation is.

So. So what you're saying is if we had a theory where the agent was able to like expand its state space through structure and parameter learning in particular, then we would have what you're looking for.

Am I misunderstanding?

Am I misunderstanding?

Am I misunderstanding?

Am I misunderstanding?

Am I misunderstanding?

Am I misunderstanding?

Am I misunderstanding?

Am I misunderstanding?

and how they unfold.

I went to light on the constraint ontology

because it was not the place for that.

But we need to look at specifically constraints

over what is observable and how they are individuated

and how they produce other constraints, et cetera, et cetera.

We have to get back at...

But this...

Constraints are dual to priors.

I mean, you're talking about the same thing.

But the formulation that is taken in physics often by belief
is DST.

It entails a preemptive space.

And this is what needs to be...

To address the problem I want to address,
this is why it has to be overcome.

And this is likely possible
given a constraint-centric ontology.

So ontology where you do not have possibilities,
then constraints of other possibilities,
but constraints, then possibilities afforded by constraints.

And I do not know how...

The thing is that those two stories are the same story.

That's what Bayesian mechanics says.

That...

It is from within DST.

But they're both from DST.

What the...

I mean, so a free energy principle theoretic model
of the belief updating of a particle in a system
is in the strong sense of like,
is dual to you.

If I have one, I have the other, right?

Is a maximum entropy dynamical systems model
of the whole system.

As in the whole system of particles
and environment to which it's coupled
and all of that sensory motor.

Like that...

That's the cool thing.

It's the FEP is the canonically optimal dynamical systems model.

It's maximum entropy in the sense that, you know,
it is the...

It makes the least assumptions
given your current state of knowledge about the system.

You know, it's Jain's...

Jain's as in, you know, J-A-Y-N-E-S.

It's Jain's optimal in the parlance of Dalton Shafiti-Vadival.

Yes, so if you have a constraints model,
you also have an FEP model and vice versa.

If you work it out from the perspective of the FEP,
you can map back onto a constraint,
but definitionally.

If you can frame a constraint model
with no reference whatsoever
to the mechanical systems theory
or pre-existing spaces,
yes, I agree, you have solved the problem.

But the paper which you refer to
frames it within the ST.

So it has...

That's an interesting point.

Which is those that we need to get beyond
to solve this specific problem again.

I do not mean to say...

Basically, you have framed this equivalence,
I think, up to details.

Let us say up to details.

I would need to think about it actually.

But with hypotheses that are precisely those
that I want to use the equivalence to overcome.

So it is not...

I guess I'm just saying that, you know,
constraints don't get you out of the problem
that you were posing originally either.

Like both...

No.

You just...

You have another kind of layer of,
you know, pre-given static.

What I'm saying is not
we have to characterize constraints.

It is that we have to have
an ontology of constraints.

So mathematical constructs
that uses constraints
with no reference to anything else,
just constraints.

What it means, I don't know.

But if we have that,
in the sense,
while conserving the sense
that is given by M. Moreno,
and while leveraging the equivalence
between constraints and belief
that...

Well, meaning belief, whatever,
that you derived,
we have one.

But it is not something that has been done

because FEP is framed within...

Well, vanilla FEP, VFEP,

is framed within DST,

which presupposes state,

and QFEP is...

Well, does not specify

what build-up survivable is,

just specify that there are observables,

that we can make observables.

But we cannot solve this within DST

or within pure abstract quantum information theory.

It is not afforded

by the structure of these theories.

One presupposes states,

the other presupposes observers.

We have to show how those

are becoming integrated.

And the only way forward,

that I see,

is a constraint ontology.

So...

Right, but what I'm saying is just,

mathematically speaking,

if you're talking about constraints,

then there is a...

In the end,

it's probably functorial.

I mean, you know,

there is a rigorous

and systematic mapping

from what you're talking about

to what the FEP is already saying.

That's...

That's really...

Yeah, I mean,

that's really what the duality is saying.

Like, yeah,

maximizing your entropy

with respect to some set of constraints
is dual to minimizing
your free energy
with respect to some model.

It's the same thing.

If you have one,
you always have the other.

If...

How would you frame
the notion of a constraint
in the sense of
Monsieur Monteville and Moreno
with no reference
to either DST
or in related states?

Well, I mean,
in these reaction networks
that they're mapping,
they are implicitly using
some kind of space.

They are not
in the reaction networks.

It is another line of work.

Right, but like,

I'm just saying

that there is

a similar kind of like
individuation of things

that's going on

when you map out

like some kind of

constrained schematic

in that sense.

Like, yeah,

the point that you're making

sort of like

grandfathers both here,

which is,

okay, but how do you,
you know,
write down
a state-space model at all
and how do they,
how do you grow them
in particular?

But I guess
I'm just saying like,
yeah, constraints
are just as subject
to this as
to the extent
that the FEP
is subject
to your criticism,
the constraint stuff
is also for the same
or dual reasons.

Like, if you just
inverted all of the arrows
in your reasoning,
you would also be able
to apply this
to the constraint stuff
equally.

Like, that's not an exit
to what you're,
pointing out.

But I guess my,
my complimentary point
is if you think
there is a,
if you think
moving to a constraint
ontology helps us,
then that's interesting
because any,

like, constraint mapping
is also dual
to an FEP theoretic model.
So maybe the FEP
can be used
like very directly
just based on
the way that you're,
like,
if you solve the problem
in terms of constraints,
you could flip it back.
it doesn't really,
it doesn't really,
and if,
I don't know,
if you're stuck
along the way,
then you can flip it
at that point
and think,
okay, well,
does looking at,
does looking at it
from this,
like,
do I want to look
at self-organization
from the perspective
of the selves
that are self-organizing
or from the perspective
of organization
and,
you know,
the maximization
and entropy
and all that stuff,

I guess.

So if you're agreeing
that the FEP
treats an entirely
different problem,
naturally different problem,
this is not a view I hold,
this is a view I oppose
specifically.

I do agree
that physics of
and by belief
derives within the ST
a rigorous relation
between,
basically,
what are the two sides
of this,
up to details.
what I am saying
is framing as a goal
the fact to leverage
that as a way
to build
the mathematical theory
of state-space construction
and I also claim
that this
would be essentially
equivalent
to build
self-consistent
and ontologically consistent
constraint ontology
for that,
for this diagram.

I'm not sure
what you're going to get

because
it is an ambiguous
that your research
as of now
is framed
within the ST
which presupposes states
which the constraint
ontology would not
and it's ambiguous
that it's very relevant
because it,
as you said,
entails
one duality
at least in one
specific case
which is this of the ST
that is
the key problem
to address.
So I think
you're,
wait,
you may be combining
two issues
and this worry.
I mean,
so there's the worry
of like
new states arising
and there's the worry
of using
states at all.
So,
I mean,
I think
the line

is very,
okay,
so
then,
yeah,
so it's,
the problem
is not really
that,
you know,
dynamical systems
theory
assumes states.
It's just
the static
nest
of
the,
or the,
you know,
the pre-determinateness
or the pre-specifiedness.
Okay,
so,
yeah.
So,
if that's specifically
you're worried
that there is
a very fruitful
way forward,
I think,
to address
a lot of them.
my,
my,
my proposal
is a state-space

construction,
not a state-space
negationism.

To get

a,

a,

I need b,

to begin with.

I need physical

things to

be there.

Well,

to have a,

actually,

self-consistent theory,

you would need this

to work from

ex-Nilo,

to get

something

from nothing.

But it is not

the issue

at hand here.

The issue

at hand here

is how

existing

things

can unfold,

create whatever

meaning states.

So,

it is not

the notion

of states

that is

the problem.

The problem
is the fact
that we do not
have a theory
of how
states are created
and we cannot
frame it
exclusively
from a mathematical
theory that
by construction
presupposes
a space
of possibilities.

We can,
however,
leverage
results from
this theory
to generalize
it to space
construction,
which is
what I'm
proposing.

So,
but what about
the renormalization
group?

So,
if you start
really from
the idea
that you
just have
observations
and they

have a scale
of noise
associated with
them,
yeah,
you can,
I think,
I mean,
there are
principled
physics-y
kind of ways
of bootstrapping
something,
kind of
saying,
well,
you know,
some stuff
changes
so fast
relative to
my reference
scale
that it
basically
averages out
and just
contributes
kind of random
fluctuations
to what I'm
observing.
in a space
of possibilities
and you
look at
basically

way to
renalyze
it,
to look
at relevant
scales,
but maybe
it would be
afforded to
have a
pastoralism,
but it does not
entail and does not
model pastoralism.
It is a model
of how complex
things,
complex dynamics
unfold.
so it is
relevant,
it is not
what I'm
pointing at.
Right.
As we
come near the
end,
for each of us,
maybe you can
spend the most
time,
Aval,
but for each
of us,
what are our
next steps?
What

cognitive
counterfactuals
and physical
embodiments
are going to
be shaped by
this work
and discussion?

Maybe we can
have non-Aval
go first
and then
you can have
the last word.
Sound good?

Okay.

Well,
personally,
I'm very
excited to
work on
the category
theory this
year,
learning about
the recent
dissertation
of Toby
Smith.

Also,
I think it's
interesting to
bring in that
niche modification,
active
inference,
variational
ecology,

a lot of
the work
that,
as Maxwell
brought up
earlier,
had been
framed within
a multi-scale
setting,
bringing some
of these new
developments over
the last one
or two years,
like the
path-based
formalization,
renormalization
group,
environmental
process,
generative process,
generative model,
symmetries,
weak Markov
blankets,
there's just
many new
concepts and
toolkits on
the table
today in
2023 than I
think there
were one or
two or three
years ago.

So it's been a
super interesting
discussion and
it brings a lot
of topics
to our
attention that
I expect
we'll be able
to reduce our
uncertainty about
a lot in the
coming year.

And I'll pass
to Alexei
then Maxwell
and then
Athlon.

The opposite
it would be
more consistent
because Alexei
and I work
directly together
in Kairos.

Okay,
Maxwell.

Well,
I mean,
Daniel,
I would echo
your overall
assessment.

I share your
enthusiasm for
where
like everything
is at

right now.

I think

we're poised

for some very

exciting work

and a lot

of very

interesting stuff

has happened

over the

short term.

In terms of

what Avel

has presented,

I think

you know,

in addition

to what you

were describing,

one of the

things I want

to like

work out

this year

is really

the philosophy

of the

free energy

principle

or Bayesian

mechanics.

I think

up until

recently,

what most

everyone

has been

doing,

myself
included,
was just
coming at
this thing
from whatever
disciplinary
background
we had
access to
and trying
to just
make sense
of it.

But I
think that
has made
us kind
of blind
to some
of the,
and everyone
has kind
of their
blind spots.

You know,
all of these
so-called
interpretations
of the
FEP
signal to
me that
there's
actually,
yeah,
there's
philosophical

work to be
done to
kind of
understand.

Jules

Hedges put
it nicely
on Twitter,
I think he
said that
he's become
convinced of
two things.

First,
the FEP
is true
in the sense
that it's
quote-unquote
the right
way to do
it,
and second,
that it's
a lot more
like boring
than all the
hype suggests
in the sense
that it's,
yeah,
it just
follows from
the way
that a bunch
of physics
hooks up.
It's very,

very fundamental
and important,
but like,
yeah,
I mean,
there are,
it's part of
the kind of
core package
now,
and there are
new,
exciting
horizons
of like
unknown,
but,
you know,
more to the
point is
that there's
like a kind
of core
philosophy
to the
FEP
that remains
to be
developed.
So,
for example,
the duality
between the
principle of
maximum
entropy,
which,
you know,

implements a
form of like
Occam's razor
in some sense
and the free
energy principle
I think is
kind of deep
and,
but we've only
started to
think about,
you know,
the implications
that have,
and Chris
Fields' group
has been
suggesting that
there's a
similar duality
between the
FEP and
unitarity
or the
conservation of
classical
information
and,
yeah,
quantum
information
theory.
So,
that's very
exciting.
Sorry if you're
talking so

long.

Awesome.

Exciting times.

Indeed.

All right,

Alexei and

Avil with the

final words.

Yeah,

so,

I will be

short.

So,

I'm really

excited about

how these

debates will

move the

lines in

theoretical

but also

practical

philosophy

and on

our

direct

perception

of

stuff

in

life

in

general

to be a

bit

broad.

Also,

I hope that

all of this
will
make,
like,
I hope this
will make the
difference in
metaphysics and
ontology and
philosophy
and meta-ontology
also,
and naturalism,
of course,
as it seems to do
in Active
Influence Lab
and Kairos.
And Avil.
So,
there are
three things
we need to do
to set all
the dust on the
argument I
generated there.
The first is to
formalize much
better the
argument and
ontological
consistency
and posterior
realism.
Both need to
be very
clear in

their reach
and
entailments.

This is
something we'll
do with
Alexi mainly
and possibly
other group
members.

Something that
we need to
do too is
to put
a lot of
work in
working out
what these
bay things
mean.

I do not
know, I can't
tell you that
it is essentially
correct.

I'm pretty
confident that
it's essentially
correct, but I
have no idea
whatsoever what
it means
mathematically,
which is
a pretty big
problem as
far as I'm
concerned.

there will
be a lot
of work
for me
and other
members,
namely
Yuna Brennack
and
Vaxim
Chimolet
to work out
the specifics
of this
mathematical
thing that
allows
order to
unfold from
chaos.
and most
pressingly, in
my opinion,
at least, it's
what I'm
supposed to
focus on in
terms of
unexpected
workflow.
It is to
work out what
this means,
how social
constraints are
enacted and
how they
evolve.

Because
ultimately, the
rest is
abstraction.

This is
what has
content.

This is what
allows us to
say, okay,
these are what
exist.

These are the
constraints that
shape things.

And this is
what is
their flow.

It is
critical from
a scientific
perspective,
because it is
what gives
me to this
entire account.

It is
critical from
a pragmatic
perspective,
because it
is what
tells us
what world
we are in
and what
is its
flow, which

is the
goal of
Kairos.
Something we
will do
with
namely
Manon
Job and
Reynolds
Gonzalo.
And so
basically,
we will
work around
the specifics
of this.
And we
will, if
someone is
interested to
collaborate,
please write
us.
We will be
very interested
because there
is a lot
of, all
of these
problems are
big problems,
all we
can say to
a lot of
firepower.
So yeah,
basically,

anyone who
wants to,
who's interested
in working
on the
specifics with
us is very
much invited
to reach
out.

Excellent.

Thank you,

Alexi,

Apple,

and

Maxwell

for

joining.

Thanks

everybody

for

watching.

Till

34.2.

All right,

it's over.

Thanks.

Yeah,

thanks both

fellows.

Um,

um,

um,

to

Thank you.

Thank you.